BUILT WITH CLAUDE: LIFE SCIENCES · BUILDER TRACK

PhenoFit

Does a reported variant actually explain *this* patient?

Prepared for Matt Deardorff, clinical geneticist · July 2026 · how it works, a worked demo case, and how it maps to the judging

The one idea

Standard variant tools run **forward**: "given a phenotype, find variants that might fit." PhenoFit runs it **backward**: the variants are already decided (the lab reported them), and the question is *"for each of these reported variants, does its gene actually explain THIS patient?"* That's the clinician's job, after the lab is done — and the part tools skip is being honest about **what's left unexplained**, because leftover features are how a second diagnosis hides.

The flow (three stages, each in its own file)

Stage 1 — Ingestion (the only place Claude is used): `extract.py` + `llm.py`

A dropped lab-report PDF (or pasted notes) goes to Claude, which **proposes** two things: the reported variants (gene + HGVS + the lab's classification) and short phenotype phrases. It uses the Anthropic SDK's structured outputs, so Claude's reply is forced into a Pydantic schema — it can't come back as prose you have to regex-scrape. Crucially, **Claude never emits an HPO id.** It only proposes phrases like "lens dislocation."

Stage 2 — Grounding (deterministic, no AI): `hpo.py` → `resolve_term()`

Every phrase Claude proposed is looked up against the real HPO ontology via the Jax API. "Lens dislocation" resolves to `HP:0001083 Ectopia lentis`. Anything that can't be grounded is reported as skipped, not guessed. So the model *drafts*, the ontology *validates* — the engine only ever sees real terms.

Stage 3 — Scoring (deterministic, sourced): `engine.py` → `review_causality()`

For each variant's gene, it fetches that gene's known disease phenotypes from HPO (`fetch_gene_phenotypes`), then checks each patient feature against them, ranks the variants, and builds the flags.

A concrete trace — the demo case

Patient has 7 features: seizure, developmental delay, gait ataxia, ectopia lentis, aortic root aneurysm, tall stature, arachnodactyly. Lab reported 3 variants: SCN1A, FBN1, MYH7.

Matching (`_match_feature`) works two ways. An **exact** match is when the gene is annotated to the patient's precise HPO term. An **ancestor** match ("via broader") uses HPO's `is_a` graph — if the patient has "gait ataxia" and the gene is annotated to the broader "gait disturbance," that still counts, but it's marked as broadened, not exact. There's a deliberate guard here: extremely broad "container" category nodes like "Neurodevelopmental abnormality" are *excluded* from ancestor matching, so a connective-tissue gene can't spuriously "explain" developmental delay through a shared grandparent.

Scoring (`_score_one`) is not a plain count — it's **rarity-weighted**. Each feature gets a weight from its information content: `weight = 0.25 + 0.75 · IC/IC_max`, where `IC = −log(fraction of all HPO diseases annotated with that term)`. Ectopia lentis (~73 diseases) is rare → high weight; seizure (~3,100 diseases) is common → low weight. The score is `sum of explained feature-weights / sum of all feature-weights`.

So for the demo:

- **FBN1 (Marfan)** explains ectopia lentis + aortic root + tall stature + arachnodactyly — the *rare, specific* findings. It doesn't touch seizures / DD / ataxia. Weighted score ≈ **66%** → ranks #1.
- **SCN1A (Dravet)** explains seizure + DD + ataxia — all *common*. Even though it's 3 features, they're low-weight, so ≈ **34%** → #2.
- **MYH7** explains only ataxia via a broad term → ≈ **14%** → #3, and its lab call was VUS anyway.

Flags (`_build_flags`) are the honest part. No single variant hits 100%, so it raises: (1) "no single variant explains everything, best is FBN1 at 4/7," and (2) the **dual-diagnosis flag** — SCN1A explains features FBN1 can't, and together they cover the whole picture, which is the ~5% "two independent causes" case. If any feature were explained by *nothing*, it'd add a residual flag pointing at genome re-analysis. If a gene's knowledge couldn't be fetched, it's marked **unscored** rather than scored zero.

The tiers themselves: 100% = Best fit, ≥60% = Possible, ≥30% = Partial, >0 = Weak, 0 = Unlikely (`_TIER_THRESHOLDS` in `engine.py`).

Why this maps to the judging

- **Impact:** it's Matt's real daily task, and because it's all HPO ids + public gene knowledge (no PHI), the same engine runs behind a hospital firewall unchanged.

- **Claude use:** the LLM is fenced to the ingestion edge with a hard contract — it proposes, the ontology disposes — so AI never touches the scored decision. That restraint is the interesting bit, not a bolted-on chatbot.
- **Depth:** ontology-aware matching, container-node exclusion, rarity weighting from the annotation network, abstention, 26 offline tests, and an eval (top-1 89%) with an honest caveat that it's a ranking check, not clinical validation.
- **Demo:** one PDF drop → ranked, cited answer with a genuine dual-diagnosis story.

Appendix — the working tool

Screenshots from the live app. A lab-report PDF is dropped in; the reported variants and the patient's phenotype are read off the page and grounded in HPO; the review then ranks which variant explains this patient and flags the rest.

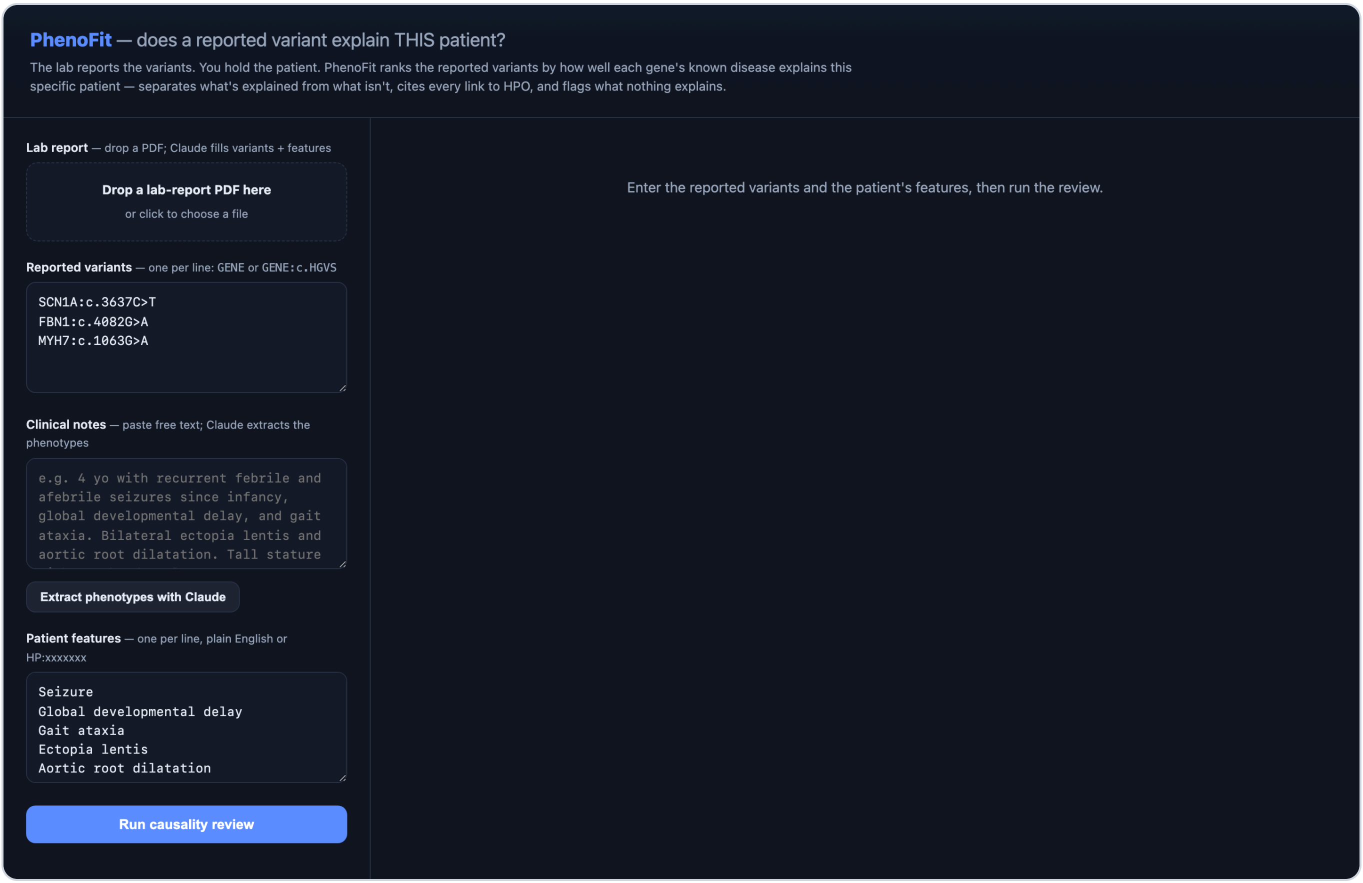

Figure 1 — Landing view. Drop a lab-report PDF, or type the reported variants and patient features directly. The AI ingestion panels appear only when an Anthropic key is configured.

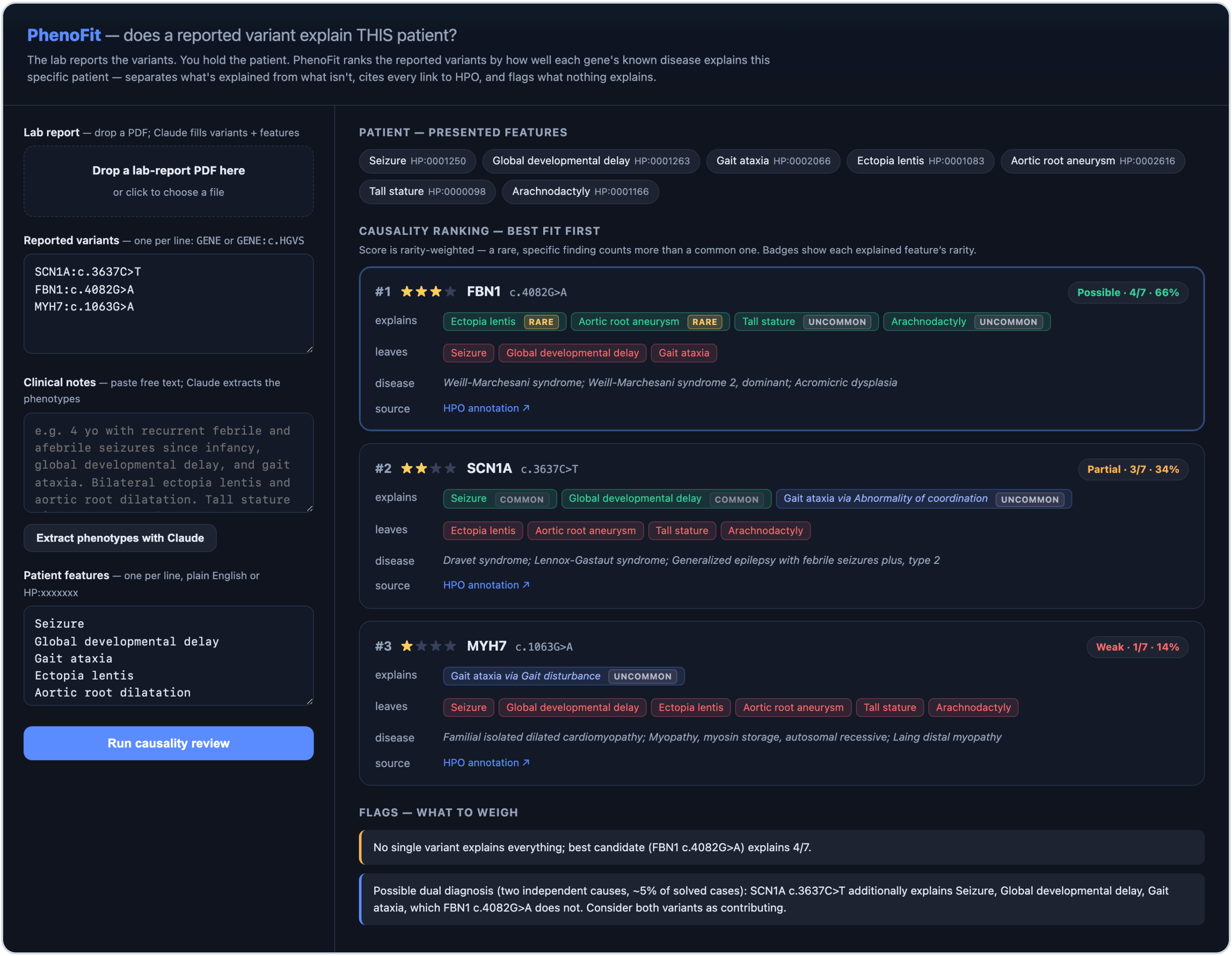

Figure 2 — The causality ranking. No single variant explains everything: FBN1 (Marfan) is the best single fit at a rarity-weighted 66% — driven by the rare, specific findings (ectopia lentis, aortic root) — while SCN1A (Dravet) explains the common seizures / developmental delay / ataxia. The dual-diagnosis flag calls out that together they cover the full picture (the ~5% two-independent-causes case).

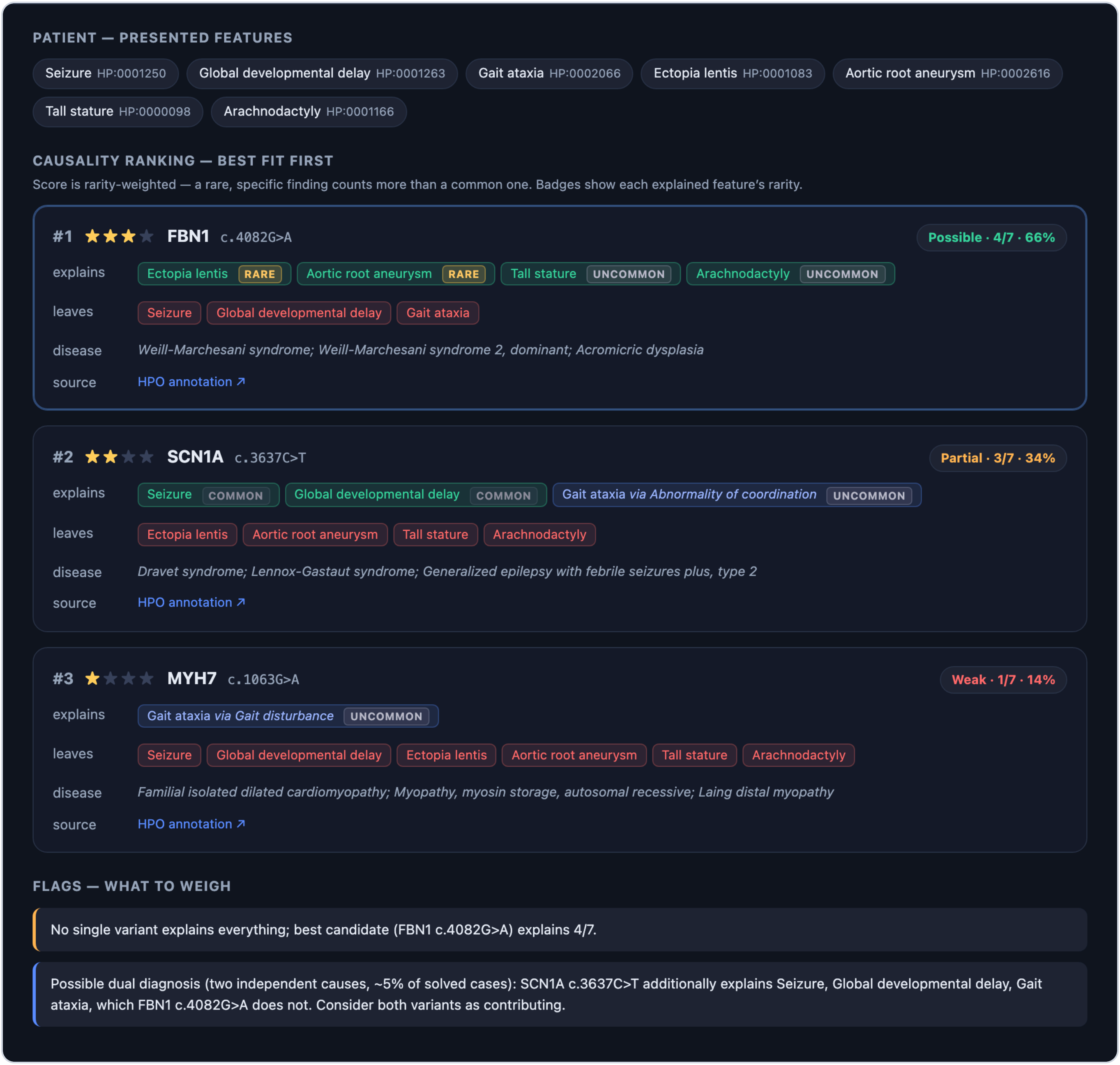

Figure 3 — Close-up of the ranked cards and flags: per-variant explained vs unexplained features, each tagged rare / uncommon / common; disease context; an openable HPO source link; and the case-level flags.